Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 3

Title: **Business Intelligence Analysis of Coffee Shop Sales**

Submitted By:

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GitHub Link: https://github.com/Amaraiah11/Business_Project_Analaysis-/tree/Amaraiah-C

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Problem Statement:

A coffee shop generates a large amount of sales data containing transaction details, product information, sales quantity, prices, store locations, and dates. Analysing this data manually can make it difficult to identify sales trends, customer preferences, and high-performing products.

The objective of this project is to use **Microsoft Power BI** to analyse coffee shop sales data and develop an interactive dashboard. The dashboard provides a centralized view of sales performance, product categories, store locations, transaction quantities, and monthly sales trends to support data-driven business decisions.

Methodology / Steps Followed

Step 1: Data Collection

The Coffee Shop Sales dataset was obtained in Excel format. The dataset contains transaction-level information about products, quantities, prices, dates, times, and store locations.

The Excel file was examined to understand the structure of the dataset and identify the fields required for analysis.

Step 2: Data Import into Power BI

- Opened **Power BI Desktop**.
- Selected the **Get Data** option.
- Imported the Coffee Shop Sales Excel file.
- Loaded the transaction data into Power BI.
- Verified the columns and data types after loading the dataset.

Step 3: Data Cleaning and Transformation

The dataset was prepared using **Power Query** before developing the dashboard.

The following activities were performed:

- Checked the dataset for missing or invalid values.
- Removed unnecessary or duplicate records where required.
- Renamed and organized fields appropriately.
- Converted date and time fields into suitable data types.
- Verified numerical fields such as quantity and unit price.
- Prepared the data for visualization and analysis.
- Created the required calculated values for sales analysis.

Step 4: Data Analysis and Calculations

A sales value was calculated using the transaction quantity and unit price.

Sales = Transaction Quantity × Unit Price

The calculated sales value was used to analyse overall revenue, product categories, product types, store locations, and monthly performance.

Key performance indicators were created to provide a quick overview of the business.

The main KPIs include:

- **Total Sales:** ₹698,812.33
- **Total Quantity Sold:** 214,470
- **Total Transactions:** 149,116

Step 5: Dashboard Development

An interactive Power BI dashboard was developed to present the analysed information in a simple and understandable manner.

The dashboard includes:

- KPI cards for overall sales performance.

- Sales analysis by product category.
- Sales analysis by product type.
- Store location performance.
- Monthly sales trends.
- Transaction quantity analysis.
- Interactive slicers and filters.
- Charts and visualizations for comparing business performance.

Key Business Insights

1. Overall Sales Performance

The coffee shop generated total sales of approximately **₹698,812.33** during the six-month period from January to June 2023.

A total of **214,470 products** were sold across **149,116 transactions**, showing a high volume of customer transactions.

2. Sales by Product Category

The **Coffee** category generated the highest sales, with approximately **₹269,952.45**.

The second-highest category was **Tea**, generating approximately **₹196,405.95**.

Other important categories included:

- Bakery – ₹82,315.64
- Drinking Chocolate – ₹72,416.00
- Coffee Beans – ₹40,085.25
- Branded Products – ₹13,607.00
- Loose Tea – ₹11,213.60
- Flavours – ₹8,408.80
- Packaged Chocolate – ₹4,407.64

This indicates that coffee and tea products contribute significantly to the overall sales of the business.

3. Sales by Store Location

The dataset contains three major store locations:

- **Hell's Kitchen**
- **Astoria**
- **Lower Manhattan**

Hell's Kitchen recorded **50,735 transactions**, followed by Astoria with **50,599 transactions** and Lower Manhattan with **47,782 transactions**.

The dashboard can be used to compare the performance of these locations and identify areas requiring further business attention.

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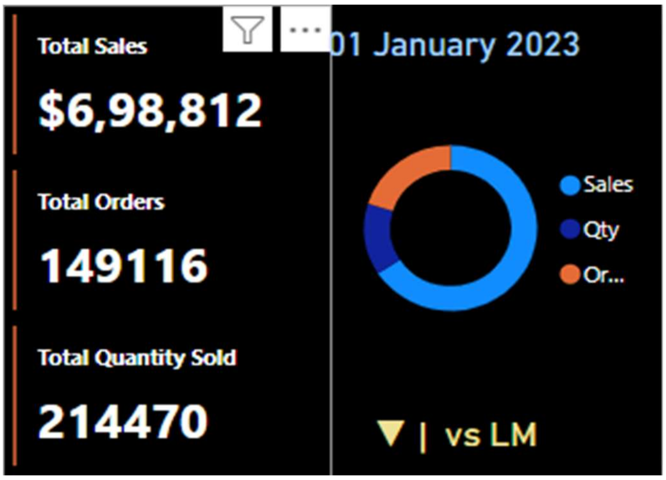
The final Power BI dashboard provides an interactive overview of the coffee shop's business performance.

The dashboard presents:

- Total Sales
- Total Quantity Sold
- Total Transactions
- Sales by Product Category
- Sales by Product Type
- Store Location Analysis
- Monthly Sales Trends
- Product Performance
- Interactive Filters and Slicers

The dashboard allows users to filter and analyse the data according to different categories, dates, locations, and products.



Conclusion:

The Coffee Shop Sales Analysis project successfully demonstrated the use of **Power BI for business data analysis and visualization**. The raw Excel dataset was imported, cleaned, transformed, and analysed to generate meaningful business insights.

The developed dashboard provides a centralized view of sales performance and helps identify high-performing product categories, products, store locations, and sales periods.

The analysis showed that **Coffee was the highest-performing product category**, while **June 2023 recorded the highest monthly sales**. Such insights can help coffee shop management understand business performance and make informed decisions regarding products, sales strategies, and store operations.